# Transformations of Absolute-Value Functions

*Vertex form, sketching the graph, reading solutions off the graph, and writing the equation from key features*

### Directions

- Every function on this sheet is built from the parent function $f(x) = |x|$ by shifts, reflections, and stretches. Vertex form is $g(x) = a\,|x - h| + k$.
- Show the work that gets you to each answer, then write the final answer on the answer line.
- On sketching problems, plot the vertex first, then one point on each arm of the V.

**1.** Give the vertex of $g(x) = |x - 4| + 1$, and state the equation of its axis of symmetry.

Answer: ________________

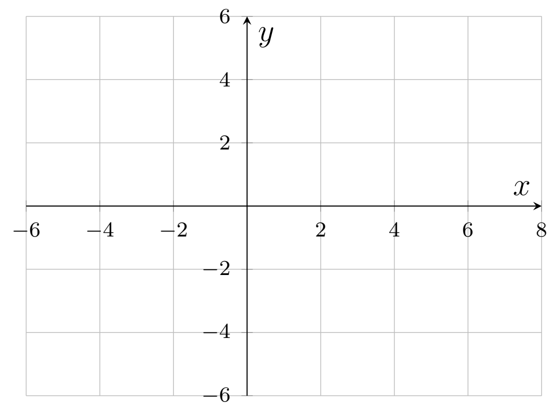
**2.** Describe in words how the graph of $g(x) = |x + 3| - 5$ is obtained from the graph of $f(x) = |x|$, then give the vertex of $g$.

Answer: ________________

**3.** For $h(x) = -2\,|x-1| + 6$: give the vertex, say whether the V opens up or down, and say whether the graph is narrower or wider than $y = |x|$.

Answer: ________________

**4.** Sketch $g(x) = |x-2| - 3$ on the grid below. Label the vertex and both $x$-intercepts, then write the vertex and the intercepts on the answer line.

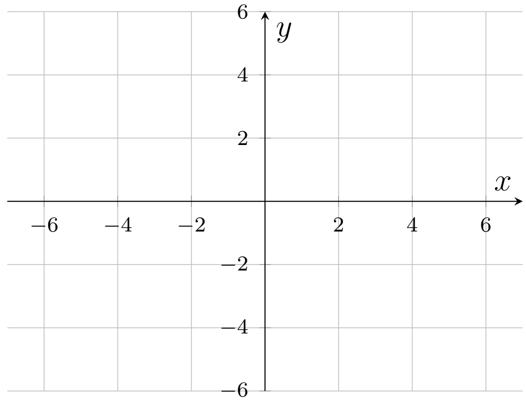



Answer: ________________

**5.** The graph of $f(x) = |x+1| - 4$ is a V with its vertex 4 units below the $x$-axis. Without graphing, find every $x$ for which $f(x) = 0$.

Answer: ________________

**6.** Sketch $g(x) = -|x| + 4$ on the grid below. Label the vertex and both $x$-intercepts, then write the vertex and the intercepts on the answer line.




Answer: __________________

**7.** Let $g(x) = 3\,|x+2| - 7$. Give the vertex, the equation of the axis of symmetry, and the value of $g(1)$.

Answer: __________________

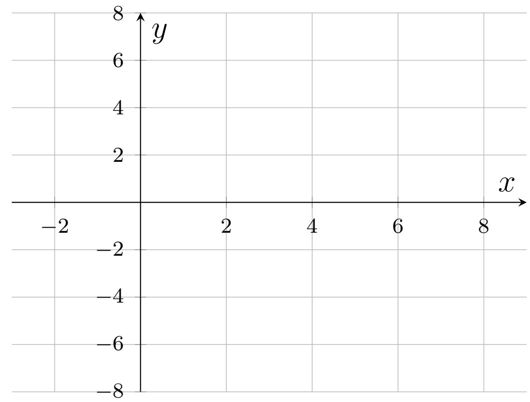
**8.** The graph of $h(x) = 2\,|x-3|-8$ is a V opening up with vertex $(3, -8)$. Use that picture to find every $x$ with $h(x) = 0$, and then every $x$ with $h(x) = 4$.

Answer: ________________

**9.** An absolute-value function $g(x) = a\,|x-h| + k$ has vertex $(5, -2)$ and its graph passes through the point $(7, 4)$. Find $a$ and $k$, then write $g(x)$ in vertex form.

Answer: ________________

**10.** On the grid below, sketch $g(x) = -2\,|x-4| + 6$. Label the vertex, both $x$-intercepts, and the $y$-intercept, then list the transformations that carry $y = |x|$ to $g$.




Answer: ________________

**11.** The graph of an absolute-value function has vertex $(-3, 8)$ and passes through $(1, 0)$. Write the function in vertex form $g(x) = a\,|x-h| + k$.

Answer: ________________

**12.** A drone flies straight along a marked trail. Its distance from the trailhead, in kilometres, after $t$ minutes is $d(t) = |2t - 6| + 1$.

(a) Rewrite $d(t)$ in the vertex form $a\,|t - h| + k$ and give the vertex.

(b) Explain what the vertex means for the drone.

(c) Find every time $t$ at which the drone is 5 kilometres from the trailhead.

Answer: ________________